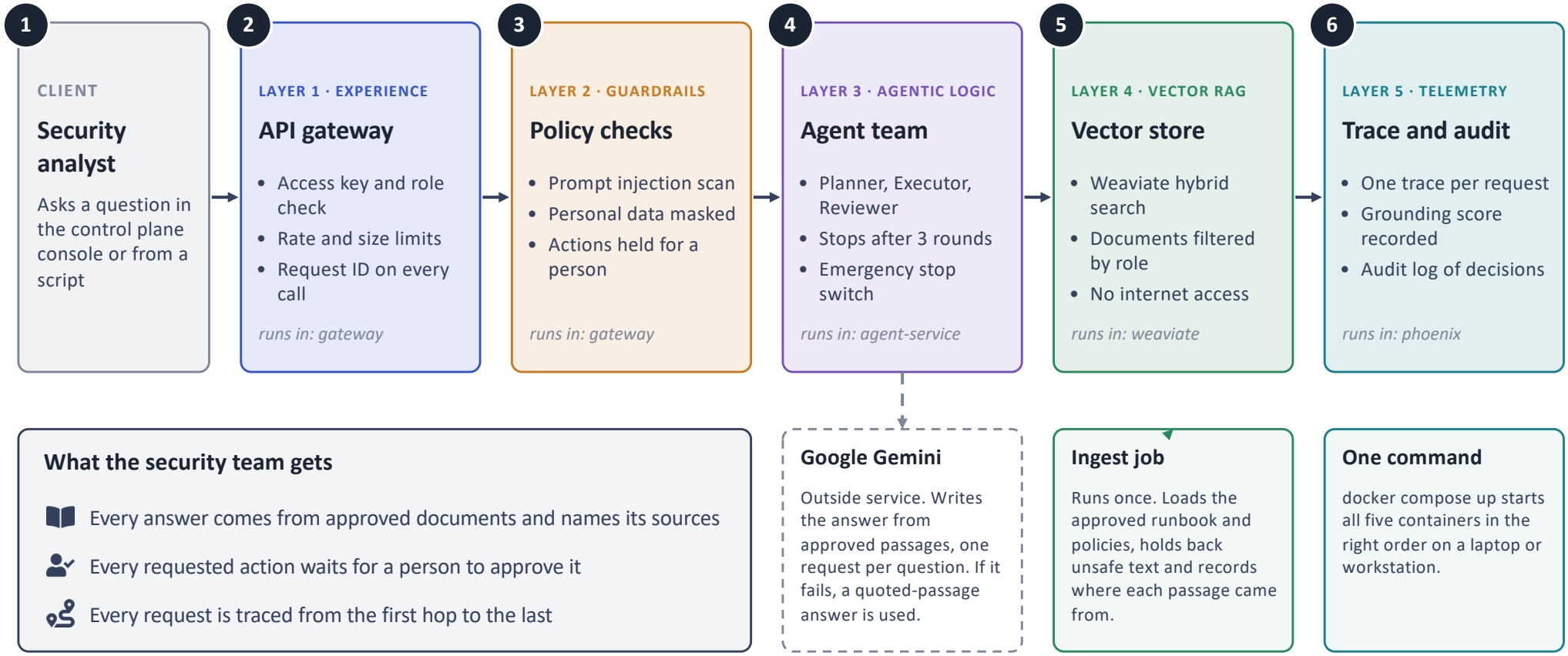
System Architecture

Governed security operations assistant: five layers and one controlled path for every request

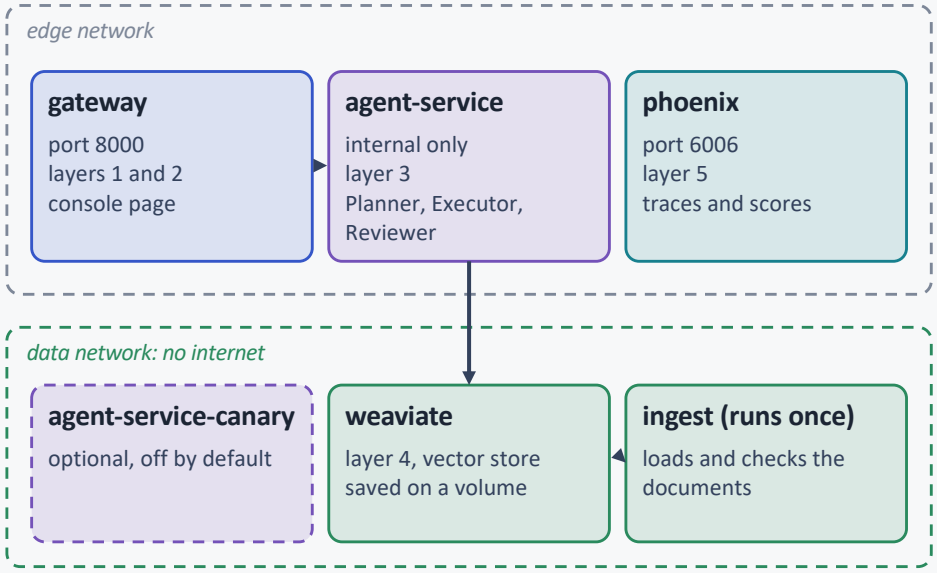


Solid arrows: the request path. Dashed arrow: optional outside call. Numbers: the six hops proved by the end-to-end check.

Deployment view

Docker Compose for building and staging, Minikube for the staged release and rollback

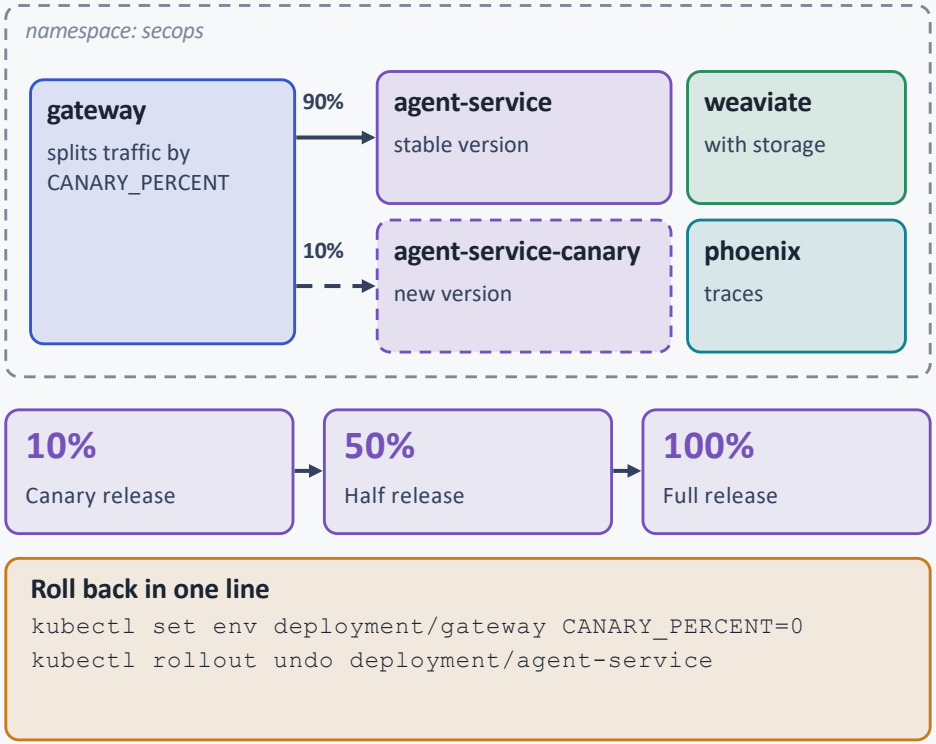
Docker Compose on a laptop or workstation



Start order: weaviate and phoenix, then the ingest job, then agent-service, then gateway. Each waits for the one before it to report healthy.

Exposure: only ports 8000 and 6006 are opened and only to this machine. Weaviate and Phoenix keep their data on named volumes.

Minikube (Kubernetes) for the staged release



Roll back in one line

```
kubectl set env deployment/gateway CANARY_PERCENT=0
kubectl rollout undo deployment/agent-service
```

One platform from four projects

Each week built one part of the platform; the capstone joins them and answers the incident theme of each week

WEEK 1

NASA Space Gallery pipeline

Incident theme

Supply chain: packages and base image were not checked against fingerprints

In the capstone

- Containers, tests and a Kubernetes rolling update
- Added now: image scan and fingerprint-checked packages

WEEK 2

Guardrailed multi-agent system

Incident theme

Rogue agent: Alibaba ROME agent broke its limits and misused computing power

In the capstone

- Planner, Executor and Reviewer
- Stop after 3 rounds and an emergency stop

WEEK 3

Monitored RAG service

Incident theme

Injection or outage: cloud outages and prompt injection hidden in documents

In the capstone

- Weaviate hybrid search and injection scan
- Phoenix tracing, scores and fallback answers

WEEK 4

Governed platform

Business problem

Most AI pilots stall at the security review stage

In the capstone

- Gateway, policy checks and human approval
- Docker Compose and a design document for the board

Together they form the five layers, packaged and released with the Week 1 practices

Layer 1 Experience	Layer 2 Guardrails	Layer 3 Agentic logic	Layer 4 Vector search	Layer 5 Telemetry	Packaging and release
Week 4	Weeks 3 and 4	Week 2	Week 3	Week 3	Week 1